

Samaneh Mohammadi

Edinburgh, United Kingdom | (+44) 7345 975267 | samaneh.mohammadi@ed.ac.uk
[LinkedIn](#) | [Personal Page](#) | [GitHub](#) | [Google Scholar](#)

WORK EXPERIENCE

Associate Researcher (Part-time)

Sep 2026 – Present

[School of Informatics, The University of Edinburgh](#) · *Edinburgh, United Kingdom*

- **Trustworthiness in LLM-based systems for high-stakes domains:** LLM-based systems are increasingly deployed in **high-stakes domains** such as hiring pipelines, where their decisions directly affect people. This project studies how **trustworthy** these systems are in such settings with a focus on their **fairness** and **security**. It is funded by and carried out in collaboration with Amazon (ongoing).

PhD Researcher (Full-time)

Mar 2026 – Present

[School of Informatics, The University of Edinburgh](#) · *Edinburgh, United Kingdom*

- **Bias and fairness in large language models (LLMs):** **Bias** in an **LLM** is not spread evenly across the network but **concentrated in particular components**. This work first locates which attention heads and feed-forward layers carry it, then fine-tunes only those components against a fairness-aware training objective (ongoing).
- **Privacy–fairness tensions in distributed ML:** differential privacy often widens performance gaps across demographic groups, creating fairness issues. To address this, we designed a **group-aware model architecture** that reduces performance gaps across groups under differential privacy, without sharing group information. Publication under review.

PhD Researcher & Developer (Full-time)

Sep 2021 – Feb 2026

[RISE Research Institutes of Sweden](#) · *Västerås, Sweden*

Conducted research and development across two EU-funded projects:

[DAIS](#) (**Distributed Artificial Intelligent Systems**) projects (EU flagship project, 47 partners):

- **Private machine unlearning framework:** letting participants privately remove their data's influence from a collaboratively trained model, removing the reliance on a central server. Publication ([CIKM '25](#)).
- **Scalable and efficient privacy preservation in distributed ML:** where a model is trained collaboratively across many devices and only model parameters are shared for aggregation, we proposed **efficient homomorphic and functional encryption frameworks** that encrypt each device's model updates locally before they are shared, while remaining practical at scale. Publications ([PMJ '25](#), [Workshop at NeurIPS '24](#), [ISPEC '23](#)).
- **Real-world model deployment:** built a physical testbed and used it to train and deploy speech emotion recognition models collaboratively across heterogeneous edge devices, measuring efficiency, fairness, and privacy trade-offs under real deployment conditions. Publications ([IJCNN '25](#), [FedCSIS '23](#)).

[DADAP](#) (**Digitizing and Automating the Diagnostic Psychiatric Assessment Process**) projects (hospital and clinical partners working on mental health disorders):

- **Healthcare AI on sensitive data:** benchmarked a range of ML models, including recent transformer architectures, on clinical mental-health tabular records collected in Swedish hospitals, and provided a broad evaluation of predictive performance and of how that performance varies across demographic groups, in order to give clinicians reliable decision support. Publication ([PharML at ECML PKDD '26](#)).

Technology Engineer / Software Developer (Full-time)

Sep 2020 – Aug 2021

[Openinside Co. W.L.L.](#) · *Manama, Bahrain (Remote)*

- **Enterprise software development (ERP):** built new Odoo modules end to end (Python backend, JavaScript front end), supporting finance and supply-chain operations for business clients.
- **Systems integration:** connected those modules to clients existing finance and supply-chain systems, keeping data and processes consistent across both.

Research Assistant (MSc Thesis)

Sep 2017 – Oct 2020

[CoinLab, University of Tehran](#) · *Tehran, Iran*

- **Anomaly detection in information networks:** developed graph-based methods that flag suspicious nodes by tracking their activity and how it changes over time, an approach applicable to fraud detection, security monitoring, and abuse detection in social networks.

SELECTED PUBLICATIONS (200+ CITATIONS)

- **Mohammadi, S.***, Tsouvalas, V.* et al., [EFU: Enforcing Federated Unlearning via Functional Encryption](#), the 34th ACM International Conference on Information and Knowledge Management (CIKM '25).
- Tsouvalas, V.*, **Mohammadi, S.*** et al., [EncCluster: Scalable Functional Encryption through Weight Clustering and Probabilistic Filters](#), Pervasive and Mobile Computing Journal (PMCJ '25).
- **Mohammadi, S.** et al., [Empirical Analysis of Asynchronous Federated Learning on Heterogeneous Devices: Efficiency, Fairness, and Privacy Trade-offs](#), International Joint Conference on Neural Networks (IJCNN '25).
- Tsouvalas, V.*, **Mohammadi, S.*** et al., [Bringing Functional Encryption to Federated Foundation Models](#), Workshop at NeurIPS '24.
- **Mohammadi, S.** et al., [Balancing Privacy and Performance in Federated Learning: A Systematic Literature Review on Methods and Metrics](#), Journal of Parallel and Distributed Computing (JPDC '24).
- **Mohammadi, S.** et al., [Balancing Privacy and Accuracy in Federated Learning for Speech Emotion Recognition](#), 18th conference on computer science and intelligence systems (FedCSIS '23), Professor Zdzisław Pawlak Award.

* Equal contribution.

EDUCATION

Ph.D. in Cyber Security, Privacy & Trust

Sep 2021 – Expected Dec 2026

[School of Informatics, University of Edinburgh](#) · *United Kingdom*

Thesis: Reconciling Privacy with Fairness and Accountability in Federated Learning under Practical Constraints. Supervisor: [Dr. Marc Juarez](#).

Note: Started Sep 2021 at RISE Research Institutes of Sweden; transferred to the University of Edinburgh, where I will complete my degree.

M.Sc. in Information Technology Engineering

Sep 2017 – Oct 2020

[University of Tehran](#) · *Iran*

Thesis: Anomaly Detection in Dynamic Information Networks. Grade: A. Supervisor: [Dr. Mostafa Salehi](#).

HONOURS & AWARDS

- **PhD Scholarship**, School of Informatics, University of Edinburgh, 2026.
- **ACM SIG Travel Grant** to present at CIKM 2025, Seoul, 2025.
- **Professor Zdzisław Pawlak Award**, FedCSIS 2023, 2023.
- **Ranked 68th among 19,598 candidates** (top 0.4%) in the national M.Sc. entrance exam in Information Technology Engineering, Iran, 2017.

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, C++, C#, MATLAB.
- **Frameworks & Libraries:** PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, Flower.
- **Machine Learning:** Federated, distributed, and on-device learning; LLM fine-tuning (LoRA / PEFT); transformer models for language, vision, and tabular data (e.g. LLaMA, BERT, ViT).
- **Privacy & Security:** Differential privacy, homomorphic encryption, functional encryption, machine unlearning.
- **Model Evaluation:** Benchmarking ML models and LLMs for trustworthiness, including privacy leakage, security risk, and fairness, across language, vision, and tabular data.
- **Tools, Infrastructure & Databases:** Git, Docker, Linux, cloud platforms; GPU cluster computing, Slurm job scheduling (srun), PostgreSQL, MS SQL Server.
- **Governance & Compliance:** GDPR, EU AI Act.

TALKS & ACADEMIC SERVICE

- **Program Committee** at [IEEE Conference on Federated Learning Technologies and Applications \(FLTA\)](#), 2025 and 2026.
- **Program Committee** at [CFAgentic @ ICML 2025 Workshop](#).
- **Invited Speaker** at [Flower Monthly](#) for “Scalable Functional Encryption in Federated Learning,” hosted by Prof. Nicholas Lane (University of Cambridge / co-founder of Flower Labs), Sep 2024.
- **Invited Speaker** at [MegaData Summer School on Federated Machine Learning](#) for “Balancing Privacy and Performance in Federated Learning,” organiser: University of Tartu, Aug 2024.
- **Conference Presentations:** ECML PKDD '26 (Naples), CIKM '25 (Seoul), IJCNN '25 (Rome), NeurIPS '24 (Vancouver), ISPEC '23 (Copenhagen), FedCSIS '23 (Warsaw).